

Tank Battle Game Project Documentation

1. Team Member Contributions

Team Member	Contribution
Liyyu	Implemented the user interface bonus challenge, including the victory screen displayed upon successful game completion.
John	Designed and implemented the Level 4 boss, including its visual appearance, size, and unique boss characteristics.
Kaung	Developed the level progression system, enabling players to advance through Levels 1–4 based on game objectives.
VJ Daniel	Implemented the Level 4 boss movement and behavior logic, including boss navigation and combat mechanics.
Ayub	Developed Levels 1–3, including map layouts, enemy placement, and gameplay functionality for the early stages of the game.

2. Level Progression Implementation

- The game consists of four levels.
- Players must defeat all enemies in the current level to progress to the next stage.
- Enemy count and difficulty increase as the player advances.
- Level 4 contains a dedicated boss battle that serves as the final challenge.
- Upon defeating the boss, the player is presented with a victory screen.

3. Boss Implementation Details

- The boss is approximately four times larger than the player tank.
- The boss uses a unique color scheme to distinguish it from regular enemies.
- The boss has 20 HP and requires multiple hits to defeat.
- The boss fires three projectiles simultaneously:
 - Forward
 - Forward-left
 - Forward-right
- The boss is placed in a dedicated arena within Level 4.
- Custom movement behavior was implemented to make the encounter more challenging.

4. Challenges Encountered

- Balancing the boss difficulty while keeping the game fair.
- Implementing a multi-projectile firing system for the boss.
- Managing level transitions and progression between stages.
- Ensuring frame-independent movement using delta time.
- Coordinating and integrating code developed by multiple team members.

5. Lessons Learned

- Gained experience with object-oriented programming in C++.
- Improved understanding of OpenGL rendering and transformations.
- Learned how to implement game AI and boss mechanics.
- Developed skills in collaborative software development using Git and GitHub.
- Better understood the importance of modular design and teamwork during game development.